



Master 2 Internship report

Dynamical properties of the Hopfield model through
bitwise arithmetic

Bastien Dumont

Supervised by Michele Castellana and David Lacoste

Institut Curie and ESPCI, Paris

March 23 - July 17, 2026

Master de Physique fondamentale et applications, parcours Physique

Scholar year 2025 - 2026

July 3, 2026

placeholder

Abstract

placeholder

Contents

1	Introduction	5
1.1	Presentation of the host laboratories	5
1.2	Subject	5
1.3	Objectives	5
2	Implementation of the algorithm	6
2.1	The Metropolis algorithm	6
2.2	First optimization: the bitwise arithmetic	6
2.3	Second Optimization: Shared random numbers	6
2.4	Structure of the code	6
3	The Ising model	7
3.1	The model	7
3.2	Bitwise Implementation	7
3.3	Phase diagram	7
3.4	Performance test	7
4	The Spinglass model	8
4.1	The model	8
4.2	Bitwise Implementation	8
4.3	Performance test	8
5	The Hopfield model	9
5.1	The model	9
5.2	Bitwise Implementation	9
5.3	Capacity of the Hopfield network	9
5.4	Performance test	9
6	Conclusion	10

1 Introduction

1.1 Presentation of the host laboratories

This internship was conducted as part of my Master 2 at the Magistère of Fundamental Physics of Paris-Saclay Université, which I completed at Aix-Marseille Université in the “Physique Fondamentale et Applications” track. It was carried out at the “Physique des Cellules et Cancer” (PCC, UMR 168) laboratory of the Institut Curie as well as at the Gulliver laboratory (UMR 7083) of the “École Supérieure de Physique et de Chimie Industrielles” (ESPCI) in Paris, and was supervised by Michele Castellana (Institut Curie) and David Lacoste (ESPCI). Although the PCC unit is a leading biological physics laboratory, part of the Institut Curie – a pioneer institution in cancer research and treatment – it hosts a renowned theory team (of which Michele Castellana is a member) that is sometimes interested in physics-driven topics such as the one investigated in this report. The Gulliver laboratory hosts both experimentalists and theoreticians (among whom David Lacoste), working at the frontier between physical, chemical and biological topics.

1.2 Subject

In October 2024, John J. Hopfield was awarded the Nobel Prize in Physics alongside Geoffrey Hinton [1] for their pioneer work on artificial neural networks. This work laid the foundations for further development and understanding of neural networks, whether they are organic or artificial, and eventually led to the arising of Artificial Intelligence (AI).

Hopfield worked on an artificial neural network, originally invented by Shun'ichi Amari in 1972 [2]. He greatly contributed to the analysis, understanding and popularization of the network in his landmark paper *Neural networks and physical systems with emergent collective computational abilities* published in 1982 [3], and the network was eventually named after him. The network, explained in more details in subsection 5.1, is similar to the Ising and the Spinglass networks: it is composed of a set of spins – called neurons in the case of the Hopfield network – $S_i \in \{-1, 1\}$ interacting with each others.

1.3 Objectives

2 Implementation of the algorithm

2.1 The Metropolis algorithm

2.2 First optimization: the bitwise arithmetic

2.3 Second Optimization: Shared random numbers

2.4 Structure of the code

3 The Ising model

3.1 The model

3.2 Bitwise Implementation

3.3 Phase diagram

3.4 Performance test

4 The Spinglass model

4.1 The model

4.2 Bitwise Implementation

4.3 Performance test

5 The Hopfield model

5.1 The model

5.2 Bitwise Implementation

5.3 Capacity of the Hopfield network

5.4 Performance test

6 Conclusion

References

- [1] The Royal Swedish Academy of Sciences. Press release: The Nobel Prize in Physics 2024. <https://www.nobelprize.org/prizes/physics/2024/press-release/>, October 2024. Awarded to John J. Hopfield and Geoffrey Hinton “for foundational discoveries and inventions that enable machine learning with artificial neural networks”. Accessed: June 2026.
- [2] Shun-ichi Amari. Learning patterns and pattern sequences by self-organizing nets of threshold elements. *IEEE Transactions on Computers*, C-21:1197–1206, 1972.
- [3] J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982.